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Robust Econometric Inference under Incomplete Information

Research proposal – Max Weber Fellowship (24 months), Department of Economics

My research asks what can be learned when economically relevant information is incomplete, identifying assumptions are only partially credible, or economic knowledge does not justify selecting a single model. At the EUI, I propose two projects extending this agenda to probabilistic expectations surveys and intergenerational mobility with missing incomes.

Project 1: What Does Survey-Reported Uncertainty Measure?

When a respondent reports a dispersed distribution for future inflation, this could reflect a precise belief that inflation is volatile, or uncertainty about the reliability of the respondent's own information and forecast. Both interpretations can generate the same histogram, but may imply different forecast revisions as new information arrives. I aim to investigate whether the *dynamics of individual survey responses* help distinguish perceived economic risk from *metacognitive uncertainty about one's own probability judgments*.

Rossi, Sekhposyan, and Soupré (2026) study the time evolution of risk and Knightian uncertainty using professional density forecasts. My question is distinct: what can repeated *individual-level* responses reveal about the respondent's own confidence in a reported distribution? Enke and Graeber (2023) study cognitive uncertainty in beliefs and economic decisions, while Giustinelli, Manski, and Molinari (2022) show that subjective probabilities may be imprecise. Building on these insights and my work on robust Bayesian inference under partially specified priors, I will model respondents as potentially entertaining *sets of predictive distributions*, with fixed-bin histograms providing incomplete observations of their beliefs. The central identification challenge is to characterize how uncertainty about one's own judgment translates into reported distributions and their revisions over time.

An estimated time-varying inflation process, for example an unobserved-components model, will provide an external benchmark to which response changes can be compared. Using repeated responses in the ECB Consumer Expectations Survey, and accounting for the survey's rolling forecast horizon, I will investigate which restrictions on belief updating can distinguish the two mechanisms and derive the resulting identified sets. The intended contribution is a methodological framework for interpreting survey-reported uncertainty, with implications for survey design.

Project 2: Rank–Rank Regressions with Missing Income Data

Rank–rank regressions are a standard measure of intergenerational mobility. Missing parental or child incomes affect not only the ranks of missing individuals but also the population ranks

assigned to observed individuals. Complete-case regressions can therefore misrepresent the population slope.

Horowitz and Manski (1998) study nonresponse bounds; Asher, Novosad, and Rafkin (2024) study mobility with coarsely observed education. Papaioannou and Speevak (1977) bound rank correlations with missing observations, while Zeng, Adams, and Bodenham (2025) derive finite-sample bounds for Spearman's footrule. Chetverikov and Wilhelm (2026) provide complete-data rank–rank inference. I target *sharp population identification and inference under unrestricted income missingness*.

My job market paper, *Sharp Identification for Regressions with Interval-Observed and Missing Covariates*, develops inference for general conditional-moment models with incomplete continuous conditioning variables. For finite samples, feasible rank completions are jointly consistent permutations, suggesting that the discrete ordering structure of ranks may yield substantially simpler identification arguments and computational methods than those required in my job market paper. I will investigate whether this discrete structure also yields tractable *population* bounds and inference accounting for both missing incomes and estimated ranks. An intergenerational-mobility application and R package will facilitate practical use.

Work Plan, EUI Fit, and Broader Relevance

Months 1–12: Develop the survey model, preliminary bounds, and an ECB survey application; derive initial rank–rank identification and inference results. I aim to circulate both preliminary drafts for feedback. *Months 13–24:* Extend the theory, complete simulations and applications, release software, and prepare both papers for journal submission.

Barbara Rossi's work on density forecasting and macroeconomic uncertainty is closely related to Project 1; Fabrizia Mealli's research on missing data and inference connects to Project 2. Both projects would benefit from interaction with the wider econometrics community at the EUI.

Gender and diversity are relevant to both projects. In the first, metacognitive uncertainty may differ by gender. In the second, mothers' incomes may be systematically less likely to be observed than fathers', particularly in developing countries. This can distort comparisons of intergenerational mobility through mothers versus fathers. My proposed method allows to study which conclusions are robust to selective missingness.

References

Asher, S., P. Novosad, and C. Rafkin (2024). "Intergenerational Mobility in India: New Measures and Estimates across Time and Social Groups." *AEJ: Applied Economics* 16(2), 66–98.

Chetverikov, D., and D. Wilhelm (2026). "Inference for Rank–Rank Regressions." arXiv:2310.15512v6.

Enke, B., and T. Graeber (2023). "Cognitive Uncertainty." *QJE* 138(4), 2021–2067.

Giustinelli, P., C. F. Manski, and F. Molinari (2022). "Precise or Imprecise Probabilities? Evidence from Survey Response Related to Late-Onset Dementia." *JEEA* 20(1), 187–221.

Horowitz, J. L., and C. F. Manski (1998). "Censoring of Outcomes and Regressors Due to Survey Nonresponse: Identification and Estimation Using Weights and Imputations." *J. Econometrics* 84(1), 37–58.

Papaioannou, T., and T. Speevak (1977). "Rank Correlation Inequalities with Missing Data." *Communications in Statistics–Theory and Methods* 6(1), 67–72.

Rossi, B., T. Sekhposyan, and M. Soupré (2026). "Understanding the Sources of Macroeconomic Uncertainty." BSE Working Paper 920.

Zeng, Y., N. M. Adams, and D. A. Bodenham (2025). "Exact Bounds of Spearman's Footrule in the Presence of Missing Data with Applications to Independence Testing." *Electronic Journal of Statistics* 19(2), 6103–6140.